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National AI Action Plan – Request for Information

On behalf of the Office of Science and Technology Policy (OSTP), the NITRD NCO requests input on the Development of an Artificial Intelligence (AI) Action Plan ("Plan").

This Plan, as directed by a Presidential Executive Order on January 23, 2025, will define the priority policy actions needed to sustain and enhance America's AI dominance, and to ensure that unnecessarily burdensome requirements do not hamper private sector AI innovation.

Through this Request for Information (RFI), OSTP and NITRD NCO seek input from industry groups, academia, and private sector organizations – including concrete AI policy actions.

In response to the RFI on the AI Action Plan, this briefing presents practical recommendations designed to sustain and enhance America's leadership in the use of AI in healthcare.

ECRI'S Key Takeaways

- **Game-changing potential:** AI has vast potential to optimize the American healthcare system by improving efficiency, reducing costs, and guiding treatment plans based on massive datasets and health research, improving care quality and patient outcomes.

- **AI is not infallible:** It's dangerous to implement AI in healthcare settings without a rigorous testing, implementation, and monitoring protocol.

- **Strong data is essential:** AI systems are only as good as the model generated from the data on which they're trained. Shortcomings in the data used in healthcare applications could harm patients.

- **Warning against AI drift:** It's essential to continuously monitor AI performance with periodic assessments to protect against degradations of the AI model due to changes in data, environments or clinical practices.

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AI in Healthcare: Improving Efficiency & Quality

By implementing clear disclosure practices, secure testing environments, and rigorous data integrity validation, we can ensure that American AI systems are not only robust and reliable but also free from unnecessary burdens. These measures lay the groundwork for a competitive and dynamic private sector, driving forward innovation that keeps us ahead of international competitors and reinforces our leadership in artificial intelligence. Adopting the standards outlined in the recommended AI strategies below will empower America to lead the global AI revolution in healthcare, solidifying the United States' position as the foremost innovator in the field.

Unlocking AI's Game-Changing Potential in Healthcare

AI has the potential to revolutionize healthcare by enhancing diagnostic accuracy, predicting patient risks, personalizing and optimizing treatment plans, and improving healthcare accessibility, ultimately improving patient outcomes. By automating routine tasks and analyzing vast amounts of clinical data, AI can support clinicians in making faster, more informed decisions while reducing errors and administrative burdens. These benefits can reduce healthcare costs and healthcare staff burn-out. However, AI must be applied with proper safeguards.

AI Misuse Could Harm Patients

Common challenges with AI technology—such as transparency, performance degradation and privacy and security concerns—can have unique and dangerous consequences in healthcare.

High quality AI models are dependent upon the robustness of the underlying algorithms and the strength of the data they rely upon. AI algorithms trained on limited or non-representative patient datasets risk producing biased or inaccurate results, which can lead to misdiagnoses, ineffective treatments, or disparities in care. Without diverse and comprehensive data, these algorithms may fail to generalize across different populations, reinforcing existing healthcare inequalities rather than improving patient outcomes.

When AI models are based on bad data, they can increase the chances of a medical error or adverse event for a patient. Medical errors generated by AI could compromise patient safety and lead to misdiagnoses and inappropriate treatment decisions, which can cause injury or death. Staff may also have difficulty determining when adverse events are attributable to AI, making such errors harder to track.

ECRI encourages continued advancement in AI-enabled medical devices that can improve healthcare efficiencies, improve clinical outcomes, and improve healthcare accessibility, and stresses that these improvements rely on robust data and sound development practices. Ensuring these standards are met is crucial for maintaining American dominance in AI innovation.

Recommended AI Strategies

ECRI recommends the following actions in the application of AI in healthcare to improve efficiency, improve patient outcomes, decrease clinical burden, make care more accessible to rural communities, and improve patient safety. These recommendations are designed to support innovation and competitiveness in the private sector, while maintaining robust safeguards that promote accountability and trust in the American healthcare system.

Promote Secure, Innovative Testing Environments

Promote the use of controlled testing environments—often called sandboxes—to validate the performance of the highest risk AI applications (e.g., those that treat or diagnose for critical patient conditions).

- In healthcare, AI-powered applications associated with higher risk are often those that influence clinical decision-making and diagnoses.
- These testing and development spaces allow innovators to rigorously assess performance and security before full-scale deployment, ensuring that high-stakes systems work as intended.
- Implementing a sandbox testing approach is feasible, as a risk stratification framework is already in place. The International Medical Device Regulators Forum (IMDRF) risk categorization for Software as a Medical Device (SaMD) is useful for clinical teams to understand the impact of applications based on its intended use. The FDA adopted this framework in their "Software as a Medical Device (SaMD): Clinical Evaluation" guidance issued on December 8, 2017.
- The IMDRF risk framework identifies two major factors which guide determination of the risk category:
    - Significance of information provided to the healthcare decision (i.e., to treat or diagnose, to drive clinical management, or to inform clinical management)
    - State of healthcare situation or condition, which identifies the intended user, disease, or condition (i.e., critical; serious; or non-serious healthcare situations).

Require Comprehensive Data Strength Reviews

Advocate for thorough assessments of the data used during the training, tuning, and testing of AI models. By examining the datasets strengths (e.g. avoiding bias and data that does not represent the target population) and by encouraging a balanced mix of real-world and synthetic data to help avoid bias, companies can better identify and address disparities among various population groups.

Require Rigorous Deep Learning Oversight

Support robust reviews of the large databases behind deep learning models. Developing national datasets—validated to be free from inadequate data points—would offer developers a trusted, standardized foundation for training deep learning algorithms. This resource would streamline the development process by reducing the time and expense spent on data cleaning and validation, ultimately accelerating innovation. Economic benefits include lower development costs, faster time-to-market for new solutions, and enhanced investor and consumer confidence, thereby reinforcing America's leadership in the global AI race.

Standardize a Bias Assessment Process

One of the major risks of AI is its potential for bias. AI models are only as good as the data on which they are trained, and biased data will result in biased models. Establish a standardized bias assessment process for training, tuning, and testing datasets.

- While social and demographic bias is a known risk to most developers, other types of bias exist. For example:
    - The hospital information systems that the AI model acquires training data from, including those systems' workflows, may not be representative of the intended use of the device or the intended patient population.
    - Variations in breast tissue density or differences in the type of imaging system collecting the data could skew diagnostic algorithms.
- Failing to mitigate these biases could lead to misdiagnoses or ineffective treatments, which would not only compromise patient care but also result in costly failures that undermine confidence in American AI systems.
- Bias assessment for training/tuning/testing datasets can include allowing synthetic data and hybrid datasets where synthetic data rounds out real-world data to prevent bias, and interrogation of the entire dataset and reporting of bias between subpopulations.
- For deep learning models, this means an assessment of the massive databases used for training and assurance that the model is being built from unbiased data.

Require Clinical Validation

Requiring clinical validation can improve device performance, which in turn may boost consumer adoption rates and reinforce U.S. leadership in AI innovation.

- Clinical validation is often missing in FDA cleared, approved or granted AI-enabled medical devices which could be a factor contributing to poor performance of these devices clinically and consumer adoption hesitancy.

- According to Nature Medicine,' from 2016 to 2022, the number of authorizations with missing clinical validation data surpassed the numbers of retrospectively and prospectively validated devices in every year since 2016.[1]
- Foster public trust in AI-enabled Software as a Medical Device (SaMD)/Software In Medical Devices (SiMD) tools as a key strategy to drive consumer adoption and reinforce U.S. leadership in AI innovation. To achieve this, ECRI emphasizes the importance of structured evaluation processes that confirm these tools are safe and effective. Clear standards for assessing AI-enabled SaMD/SiMD can help ensure these devices deliver clinical value, especially in situations where formal oversight may be limited.
- ECRI recommends that healthcare providers and payers independently assess these tools to confirm their clinical value, especially in cases where regulatory oversight is limited. Building confidence in AI-enabled medical devices through transparent evaluation processes will encourage broader adoption and accelerate economic growth.

Require Transparency and Standardization of Disclosed Data

Encourage a consistent approach where companies share key technical details with consumers—such as through standardized "model cards" or "AI nutrition fact labels"—to clearly explain how AI systems are built and function.

- This straightforward disclosure supports market confidence and enables informed decision-making.
- Measures of recall and precision should be included.
- Enhanced transparency builds public trust and acceptance, leading to stronger global demand for innovative American AI solutions.

Require Safety Reporting

Clinical literature indicates there may be underreporting in regulatory databases and industry press. Require and consolidate AI/ML product safety reporting to address patient harm or preventable harm.

- An AI product must be regularly monitored to ensure ongoing effectiveness and safety. This may entail ensuring reliable communication with both product users and the product vendor/developer, regularly assessing the availability of alternative, improved AI products, assembling a committee to review the AI product periodically, performing

[1] Chouffani El Fassi, S., Abdullah, A., Fang, Y. *et al.* Not all AI health tools with regulatory authorization are clinically validated. *Nat Med* **30**, 2718–2720 (2024). https://doi.org/10.1038/s41591-024-03203-3, last accessed March 11, 2025

audits of the AI model's performance, maintaining a record of AI product adverse events, and monitoring event reporting in regulatory databases.

- For healthcare applications coordinated under the Assistant Secretary for Technology Policy / Office of the National Coordinator for Health Information Technology (or government agencies other than FDA), consider guiding users to perform a risk assessment and to review the AI/ML product periodically to identify drift or bias, where monitoring frequency should be defined based on intensity of use of and criticality of the application. Refer to Risk Assessment Considerations from ISO 14971.

Mitigate Effects of AI Drift Over Time

Drift occurs when an AI model's performance degrades due to changes in data, environments, or clinical practices.

- In healthcare AI applications, drift can happen as patient demographics shift, medical guidelines evolve, or new treatments emerge, causing the model to become less accurate. Without regular monitoring and updates, an AI tool that was once reliable may start making incorrect predictions, leading to potential safety risks for patients.
- To ensure AI remains safe and effective, healthcare organizations must implement continuous validation, retraining, and human oversight to detect and correct drift before it impacts care.
- Manufacturers must also maintain transparency with regulators and providers on planned and unplanned software updates.
- The current FDA recommendation allowing for a predetermined change control plan (PCCP) specifies the modifications the vendor intends to implement without further regulation in the event that a change to the AI model becomes necessary. This provision allows flexibility and innovation while maintaining accountability and building consumer trust.

Train Healthcare Professionals

Training healthcare professionals to interpret AI results critically is essential to prevent errors and ensure optimal outcomes.

- Human-AI interaction performance depends on factors like tailored integration, minimizing cognitive burden, addressing automation bias, and fostering trust through transparent communication.
- Automation bias, where users overly rely on AI-generated results without critical evaluation, is a significant concern.

Total Systems Safety

AI risk assessment processes should be part of a Total Systems Safety (TSS) approach. Healthcare organizations should examine potential systems factors that could contribute to failures in AI technologies. TSS principles are also reflected in the Health AI Partnership (HAIP) framework for risk management in AI-based clinical applications.

By adopting these recommendations and tactics, the national AI Action Plan can establish a framework that safeguards patient safety and economic growth opportunities while encouraging American innovation.

ECRI: Advancing Effective, Evidence-Based Healthcare

ECRI has been a leader in patient safety, healthcare quality, and evidence-based medicine for over 50 years. As an independent, nonprofit organization, ECRI provides trusted guidance to healthcare leaders and policymakers to improve safety, quality, and cost-effectiveness in healthcare. With the strictest conflict of interest rules in the industry, ECRI remains a leading resource for unbiased, evidence-based decision-making.

ECRI is federally certified as a Patient Safety Organization (PSO) and collaborates with over 1,800 healthcare facilities nationwide, including health systems, hospitals, ambulatory surgery centers, and aging services. By analyzing ECRI's database of over 7 million patient adverse events and "near misses," ECRI identifies the greatest threats to patient safety and care efficiency and develops evidence-based solutions.

ECRI partners with government and public agencies, payers, medical liability insurers, the financial, pharma, and biotech communities, medical device manufacturers, and healthcare providers. For federal agencies, ECRI provides evidence-based research, patient safety advisories, and policy guidance. Examples include:

- FDA: Medical device safety and regulatory insights
- CDC: Infection prevention and patient safety advisories
- VA and DoD: Patient safety initiatives for veterans and military personnel
- CMS and HRSA: Evidence-based research and policy guidance to inform healthcare programs

ECRI is the only organization that leads independent evaluations of medical devices. Each year, ECRI tests thousands of devices, empowering healthcare organizations to make informed procurement decisions. ECRI is one of the few Evidence-based Practice Centers (EPCs) designated by the U.S. Agency for Healthcare Research and Quality (AHRQ).

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